

Practice 1: Round to estimate

Name: _____ Date: _____

Round each number to the place named, then estimate.

1. Estimate $487 + 236$ by rounding each number to the nearest hundred.

Work space

2. Estimate $7,842 - 3,196$ by rounding to the nearest thousand. Explain your choices.

Work space

Practice 2: Rounding with decimals

Round first. Show the rounded numbers and the estimate.

1. Estimate $36.78 + 14.21$ to the nearest whole number.

Work space

2. Estimate 9.63×4.2 by rounding each factor to the nearest whole number.

Work space

Practice 3: Estimate products

Use a convenient place value. Write an estimate, then calculate exactly.

1. Estimate 398×21 by rounding each factor to its greatest place.

Work space

2. A theater has 48 rows with 19 seats in each row. About how many seats are there?

Work space

Practice 4: Compatible numbers

Replace numbers with nearby friendly numbers to estimate mentally.

1. Estimate $198 + 297 + 403$ using compatible numbers. Explain.

Work space

2. Estimate $4,812 \div 61$ using compatible numbers.

Work space

Practice 5: Mental addition strategies

Use compensation or place-value parts. Show your mental steps.

1. Find $399 + 268$ mentally by compensating.

Work space

2. Find $2,497 + 1,806$ mentally. Describe a strategy.

Work space

Practice 6: Mental subtraction strategies

Use compensation or count up.

1. Find $700 - 298$ mentally using compensation.

Work space

2. Find $5,000 - 2,796$ by counting up from the smaller number.

Work space

Practice 7: Mental multiplication

Break apart a factor or use a known fact.

1. Find 25×16 mentally. Show two helpful facts.

Work space

2. Find 48×5 mentally and explain why your strategy works.

Work space

Practice 8: Mental division

Use place value, factors, or a related multiplication fact.

1. Find $3,600 \div 9$ mentally.

Work space

2. Find $2,400 \div 60$ mentally. Show how you simplify.

Work space

Practice 9: Front-end estimation

Estimate using the leading digits, then decide whether your estimate is high or low.

1. Estimate $6,482 + 2,759$ by adding the thousands first. Is it high or low?

Work space

2. Estimate $9,876 - 4,321$ using front-end estimation. Is it high or low?

Work space

Practice 10: Reasonable answers

Estimate to check whether each exact answer is reasonable.

1. Maya says $612 \times 7 = 4,284$. Use an estimate to check her answer.

Work space

2. A student says $4,950 \div 5 = 99$. Is this reasonable? Prove it with an estimate or related fact.

Work space

Practice 11: Choose the operation

Estimate first, then identify the operation and solve.

1. A school buys 6 boxes of 24 pencils. About how many pencils? Give an exact answer too.

Work space

2. There are 238 guests and 97 leave. About how many remain? Give an exact answer too.

Work space

Practice 12: Two-step estimates

Estimate each step and state your assumptions.

1. A store sells 27 shirts at \$19 each and 14 hats at \$12 each. About how much money is that?

Work space

2. A farm packs 8 crates of 46 apples, then gives away 75. About how many remain?

Work space

Practice 13: Exact answer after estimating

Use estimation to plan, then calculate exactly.

1. A bus travels 186 miles Monday and 217 Tuesday. Estimate the total, then find the exact total.

Work space

2. A warehouse has 2,304 cans and ships 879. Estimate and then find how many are left.

Work space

Practice 14: Bounds and range

Use the given limits or rounding to describe a range of possible answers.

1. Each of 32 classrooms has between 24 and 29 students. Give a lower and upper bound for the school population.

Work space

2. A number rounds to 600 to the nearest hundred. What is the smallest and largest whole number it could be?

Work space

Practice 15: Draw a plan

Before calculating, list the steps and choose an operation for each.

1. Four teams each collect 38 cans. They pack them equally into 8 boxes. How many cans per box, and how many are left?

Work space

2. A \$50 budget pays for 3 notebooks at \$6 and a backpack at \$28. How much remains?

Work space

Practice 16: Tables and data

Estimate totals from the data, then calculate exactly.

1. Books read: Ana 18, Ben 24, Chen 31, Dina 27. Estimate the total to the nearest ten, then find it.

Work space

2. Daily steps: 4,820; 5,190; 4,760. Estimate the three-day total to the nearest thousand.

Work space

Practice 17: Money and decimals

Estimate money amounts sensibly, then solve exactly.

1. Three tickets cost \$14.95 each. About how much? What is the exact cost?

Work space

2. You have \$40 and spend \$7.85 and \$12.60. Estimate and find the exact amount left.

Work space

Practice 18: Explain and compare strategies

Solve each problem two ways when asked.

1. Find $999 + 1,001$ in two mental ways. Which is more efficient?

Work space

2. Estimate 51×19 two ways. Compare your estimates.

Work space

Practice 19: Mixed challenge

Show an estimate, an exact answer, and a sentence explaining reasonableness.

1. A stadium has 9 sections with 418 seats each. If 735 seats are empty, about how many are occupied?

Work space

2. A 5-day fundraiser collects \$286, \$419, \$305, \$198, and \$392. Is the total over \$1,500? Estimate and decide.

Work space

Practice 20: Strategy review

Choose a strategy, show your work, and check your answer with estimation.

1. A library has 3,248 fiction books and 1,876 nonfiction books. It donates 925 books. How many remain?

Work space

2. A factory makes 24 boxes of 36 markers and shares them equally among 9 classrooms. How many markers per classroom?

Work space